COURSE SLIDES

Microsoft Certified Azure AI Engineer Associate AI-102 Training

TGS Ref No: TGS-2023036651

TSC: Artificial Intelligence Application in Product Development (ICT-TEM-4034-1.1)

Conducted by Tertiary Infotech Academy Pte Ltd
UEN 201200696W

Version 1.0
10 Jul 2026

COURSE ADMINISTRATION

Welcome and Housekeeping

TRAQOM / SSG

Digital Attendance (Mandatory)

Take AM, PM and Assessment digital attendance for WSQ-funded courses.
Scan the SSG portal QR code shown by the trainer or administrator.
Minimum 75% attendance is required for funding and assessment eligibility.

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation and software engineering.

DELIVERS

WSQ courses on AI agents, automation (n8n) and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

YOUR TRAINER - GENERAL

About the Trainer

?

Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

ICE-BREAKER

Let's Know Each Other

Your name and organisation or role.
Your experience with Azure or AI services.
Your goal for this AI-102 course.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively - no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

SCHEDULE

Lesson Plan - 2 Days

Day 1

Planning, security, monitoring, responsible AI.
Generative AI, Foundry, Azure OpenAI, RAG.
Agentic AI and Foundry Agent Service.
Vision, video, NLP, speech, translation.

Day 2

Custom language and question answering.
Azure AI Search, knowledge mining, vector search.
Document Intelligence and Content Understanding.
Capstone review and final assessment.

WSQ ALIGNMENT

Skills Framework

TSC Title: Artificial Intelligence Application in Product Development.
TSC Code: ICT-TEM-4034-1.1.
Abilities: analyse algorithms, evaluate AI applications, assess feasibility and improvements.
Knowledge: AI applications, performance analysis, evaluation methods, algorithm design and implementation.

WHAT YOU WILL ACHIEVE

Learning Outcomes

Analyse Azure AI applications and establish their correlation with efficiency.
Identify and evaluate strengths and limitations of Microsoft Azure AI applications.
Assess feasibility and improvements when applying Azure AI to product and maintenance processes.

LAB-ALIGNED FLOW

Course Outline

Lab 01: Plan, Manage, Monitor, and Secure an Azure AI Solution
Lab 02: Responsible AI, Content Safety, Governance
Lab 03: Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG
Lab 04: Agentic AI, Foundry Agent Service, Multi-Agent Concepts
Lab 05: Computer Vision, Custom Vision, Video Insights
Lab 06: NLP, Language, Speech, Translation
Lab 07: Custom Language Models and Question Answering
Lab 08: Azure AI Search, Knowledge Mining, Vector Search
Lab 09: Document Intelligence and Content Understanding
Lab 10: AI-102 Capstone and Course Review

Assessment

Written Assessment - Short-Answer Questions (WA-SAQ).
Practical Performance (PP).
Open book assessment: slides, Learner Guide and approved materials only.
Final assessment starts on Day 2 at 4:00 PM.

ASSESSMENT RULES

Briefing for Assessment

Place phones and other materials under the table or on the floor.
No photos or recording of assessment scripts.
No discussion during assessment.
Submit assessment scripts when time is up.

SSG REQUIREMENTS

Criteria for Funding

Minimum attendance rate of 75% based on SSG Digital Attendance record.
Complete the assessment and be assessed as Competent.

RESOURCES

LMS / TMS and Labs

LMS/TMS: https://lms-tms.tertiaryinfotech.com
Course labs: GitHub labs folder for TGS-2023036651.
Use the lab sequence in this deck, Learner Guide and Lesson Plan.

AI-102 FOUNDATIONS

Day 1

Planning, responsible AI, generative AI, agents, vision and language.

LAB OVERVIEW

Lab 01: Plan, Manage, Monitor, and Secure an Azure AI Solution

Main topic: Planning and operations
Scenario: A company wants a customer support AI platform with document search, answer generation, sentiment analysis, speech transcription, and image processing. You must plan the Azure AI resources.
Validation: You should have a service map, security plan, monitoring plan, and architecture diagram.

WHAT YOU WILL DO

Lab 01: Objectives

Select appropriate Azure AI services.
Plan resource deployment and endpoints.
Identify authentication, keys, monitoring, and cost controls.
Prepare an AI solution architecture.

Lab 01: Imported Reference Diagram

Azure OpenAI

The Azure OpenAI service provides access to OpenAI generative AI models including the GPT family of large and small language models and DALL-E image-generation models within a scalable and securable cloud service on Azure.

Azure AI Vision

The Azure AI Vision service provides a set of models and APIs that you can use to implement common computer vision functionality in an application. With the AI Vision service, you can detect common objects in images, generate captions, descriptions, and tags based on image contents, and read text in images.

Azure AI Speech

The Azure AI Speech service provides APIs that you can use to implement *text to speech* and *speech to text* transformation, as well as specialized speech-based capabilities like speaker recognition and translation.

Azure AI Language

The Azure AI Language service provides models and APIs that you can use to analyze natural language text and perform tasks such as entity extraction, sentiment analysis, and summarization. The AI Language service also provides functionality to help you build conversational language models and question answering solutions.

How to use it

Connect the diagram to the lab scenario: A company wants a customer support AI platform with document search, answer generation, sentiment analysis, sp...
Use it as a reference while completing the design steps.
Imported from source PPT slide 25.

FOLLOW IN ORDER

Lab 01: Step Sequence

1. Map Requirements to Services
2. Plan Foundry Resources
3. Plan Security
4. Plan Monitoring and Cost
5. Draw Architecture

CHECK BEFORE MOVING ON

Lab 01: Validation and Checkpoints

You should have a service map, security plan, monitoring plan, and architecture diagram.
Why is service selection part of AI engineering?
Why should keys be protected?
What should be monitored in an AI service?
What is the role of responsible AI planning?

LAB OVERVIEW

Lab 02: Responsible AI, Content Safety, Governance

Main topic: Responsible AI
Scenario: The support AI assistant may receive abusive user input, produce inaccurate answers, or expose sensitive data. You must design responsible AI controls.
Validation: You should have responsible AI controls, safety policy, scenario decisions, and incident plan.

WHAT YOU WILL DO

Lab 02: Objectives

Apply responsible AI principles.
Configure content safety concepts.
Explain content filters, blocklists, prompt shields, and harm detection.
Design a responsible AI governance framework.

Lab 02: Lab Alignment Diagram

Lab-aligned design flow

Scenario	>	Plan	>	Configure / Design	>	Validate	>	Review

Scenario

The support AI assistant may receive abusive user input, produce inaccurate answers, or expose sensitive data. You must design responsible AI controls.

Lab steps

Review Responsible AI Principles
Define Content Safety Controls
Create a Safety Policy
Test Governance Scenarios
Create an Incident Plan

Validation

You should have responsible AI controls, safety policy, scenario decisions, and incident plan.

FOLLOW IN ORDER

Lab 02: Step Sequence

1. Review Responsible AI Principles
2. Define Content Safety Controls
3. Create a Safety Policy
4. Test Governance Scenarios
5. Create an Incident Plan

CHECK BEFORE MOVING ON

Lab 02: Validation and Checkpoints

You should have responsible AI controls, safety policy, scenario decisions, and incident plan.
What are content filters?
What is a prompt shield?
Why is human escalation important?
Who is accountable for AI output?

LAB OVERVIEW

Lab 03: Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG

Main topic: Generative AI
Scenario: The company wants a grounded assistant that answers questions from internal support articles and generates draft replies.
Validation: You should have model selection notes, prompt templates, prompt flow diagram, RAG design, and evaluation table.

WHAT YOU WILL DO

Lab 03: Objectives

Plan a generative AI solution with Microsoft Foundry.
Select and deploy a model conceptually.
Design prompt templates and prompt flow.
Implement a RAG pattern conceptually.
Evaluate model and flow outputs.

GENERATIVE AI

Lab 03: Imported Reference Diagram

How to use it

Connect the diagram to the lab scenario: The company wants a grounded assistant that answers questions from internal support articles and generates dra...
Use it as a reference while completing the design steps.
Imported from source PPT slide 309.

FOLLOW IN ORDER

Lab 03: Step Sequence

1. Select a Model
2. Design Prompt Templates
3. Plan Prompt Flow
4. Design RAG
5. Evaluate Outputs

CHECK BEFORE MOVING ON

Lab 03: Validation and Checkpoints

You should have model selection notes, prompt templates, prompt flow diagram, RAG design, and evaluation table.
What is RAG?
Why are prompt templates useful?
What does tracing help debug?
How can feedback improve a generative AI solution?

LAB OVERVIEW

Lab 04: Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Main topic: Agents
Scenario: The support AI assistant needs to look up order status, search support documents, draft a reply, and escalate complex cases to a human agent.
Validation: You should have agent use case, resource plan, tool table, orchestration diagram, and multi-agent notes.

WHAT YOU WILL DO

Lab 04: Objectives

Explain agent roles and use cases.
Plan Foundry Agent Service resources.
Design custom agent tools and workflows.
Review multi-agent orchestration concepts.

Lab 04: Lab Alignment Diagram

Lab-aligned design flow

Scenario	>	Plan	>	Configure / Design	>	Validate	>	Review

Scenario

The support AI assistant needs to look up order status, search support documents, draft a reply, and escalate complex cases to a human agent.

Lab steps

Define Agent Use Case
Plan Agent Resources
Design Tool Calls
Design Orchestration
Multi-Agent Review

Validation

You should have agent use case, resource plan, tool table, orchestration diagram, and multi-agent notes.

FOLLOW IN ORDER

Lab 04: Step Sequence

1. Define Agent Use Case
2. Plan Agent Resources
3. Design Tool Calls
4. Design Orchestration
5. Multi-Agent Review

CHECK BEFORE MOVING ON

Lab 04: Validation and Checkpoints

You should have agent use case, resource plan, tool table, orchestration diagram, and multi-agent notes.
What is an AI agent?
Why do agents need tool permissions?
What is orchestration?
Why should autonomous actions be constrained?

LAB OVERVIEW

Lab 05: Computer Vision, Custom Vision, Video Insights

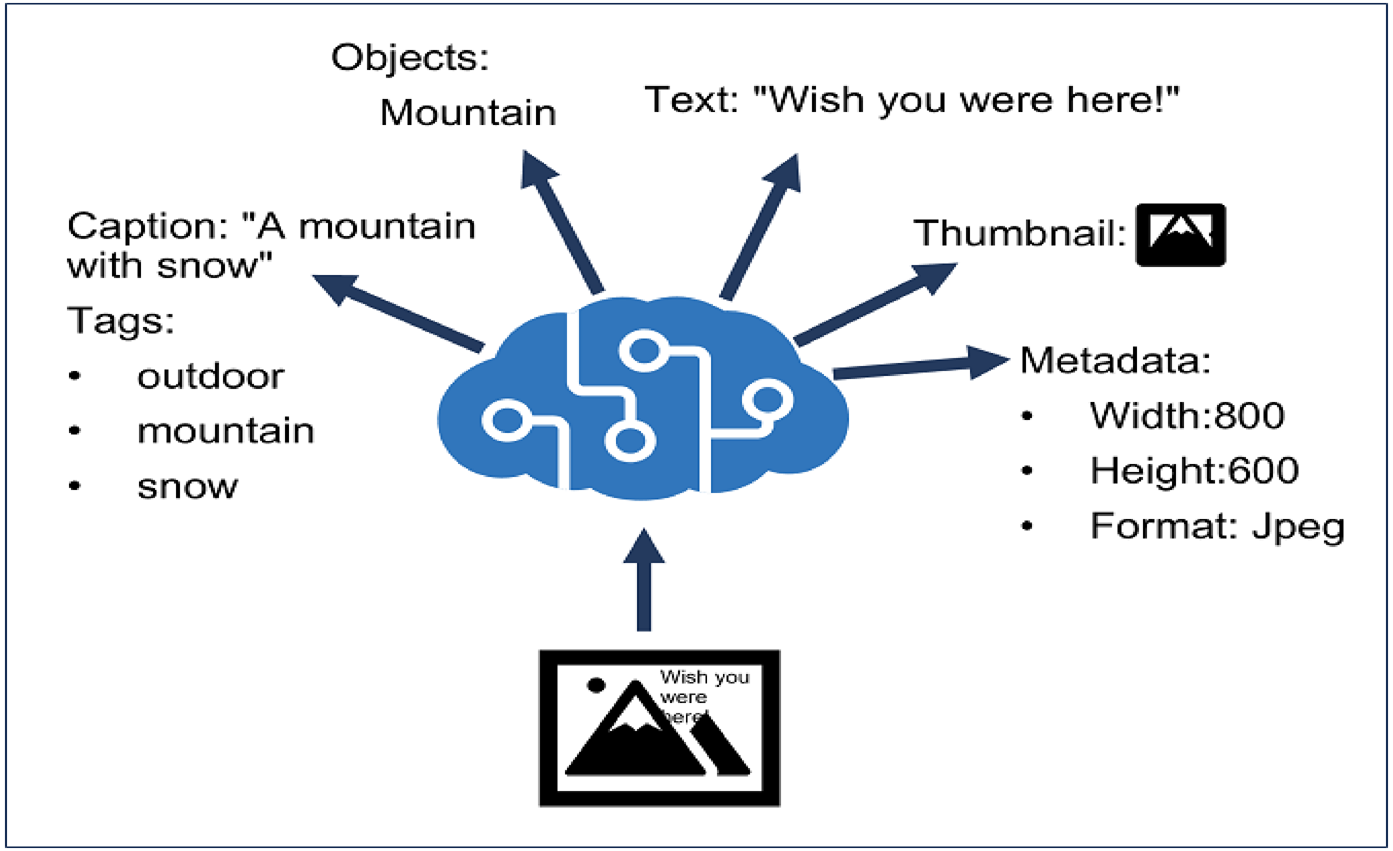
Main topic: Vision
Scenario: The company wants to inspect product photos, detect damaged items, read text from images, and extract insights from product demo videos.
Validation: You should have vision mapping, request plan, custom vision plan, video insight notes, and responsible AI review.

WHAT YOU WILL DO

Lab 05: Objectives

Analyze images with Azure AI Vision concepts.
Select visual features for image processing requirements.
Compare image classification and object detection.
Plan custom vision training.
Explain video insight scenarios.

Lab 05: Imported Reference Diagram



How to use it

Connect the diagram to the lab scenario: The company wants to inspect product photos, detect damaged items, read text from images, and extract insights...
Use it as a reference while completing the design steps.
Imported from source PPT slide 73.

FOLLOW IN ORDER

Lab 05: Step Sequence

1. Map Vision Requirements
2. Plan an Image Request
3. Custom Vision Plan
4. Video Insights Plan
5. Responsible AI Review

CHECK BEFORE MOVING ON

Lab 05: Validation and Checkpoints

You should have vision mapping, request plan, custom vision plan, video insight notes, and responsible AI review.
What is OCR?
How is object detection different from classification?
What does a custom vision model require?
Why is video analysis sensitive?

LAB OVERVIEW

Lab 06: NLP, Language, Speech, Translation

Main topic: Language and speech
Scenario: The support platform must analyze customer messages, detect PII, translate text, transcribe calls, and generate spoken responses.
Validation: You should have language task mapping, text pipeline, speech pipeline, SSML notes, and translation notes.

WHAT YOU WILL DO

Lab 06: Objectives

Analyze and translate text.
Extract key phrases, entities, PII, language, and sentiment.
Process speech with speech-to-text and text-to-speech.
Plan translation and SSML use.

Lab 06: Imported Reference Diagram

Feature	Image
Speech to text	mcr.microsoft.com/product/azure-cognitive-services/speechservices/speech-to-text/about
Custom Speech to text	mcr.microsoft.com/product/azure-cognitive-services/speechservices/custom-speech-to-text/about
Neural Text to speech	mcr.microsoft.com/product/azure-cognitive-services/speechservices/neural-text-to-speech/about
Speech language detection	mcr.microsoft.com/product/azure-cognitive-services/speechservices/language-detection/about

How to use it

Connect the diagram to the lab scenario: The support platform must analyze customer messages, detect PII, translate text, transcribe calls, and generat...
Use it as a reference while completing the design steps.
Imported from source PPT slide 61.

FOLLOW IN ORDER

Lab 06: Step Sequence

1. Map Language Tasks
2. Design Text Analysis Pipeline
3. Design Speech Pipeline
4. SSML Review
5. Translation Review

CHECK BEFORE MOVING ON

Lab 06: Validation and Checkpoints

You should have language task mapping, text pipeline, speech pipeline, SSML notes, and translation notes.
What is entity recognition?
Why detect PII before storing text?
What is SSML?
When should translation outputs be reviewed?

AI-102 INTEGRATION AND REVIEW

Day 2

Custom language, search, document intelligence, capstone and assessment.

LAB OVERVIEW

Lab 07: Custom Language Models and Question Answering

Main topic: Custom language
Scenario: The support assistant must recognize customer intents and answer frequently asked questions from approved sources.
Validation: You should have intent/entity table, training plan, question answering plan, multilingual strategy, and backup notes.

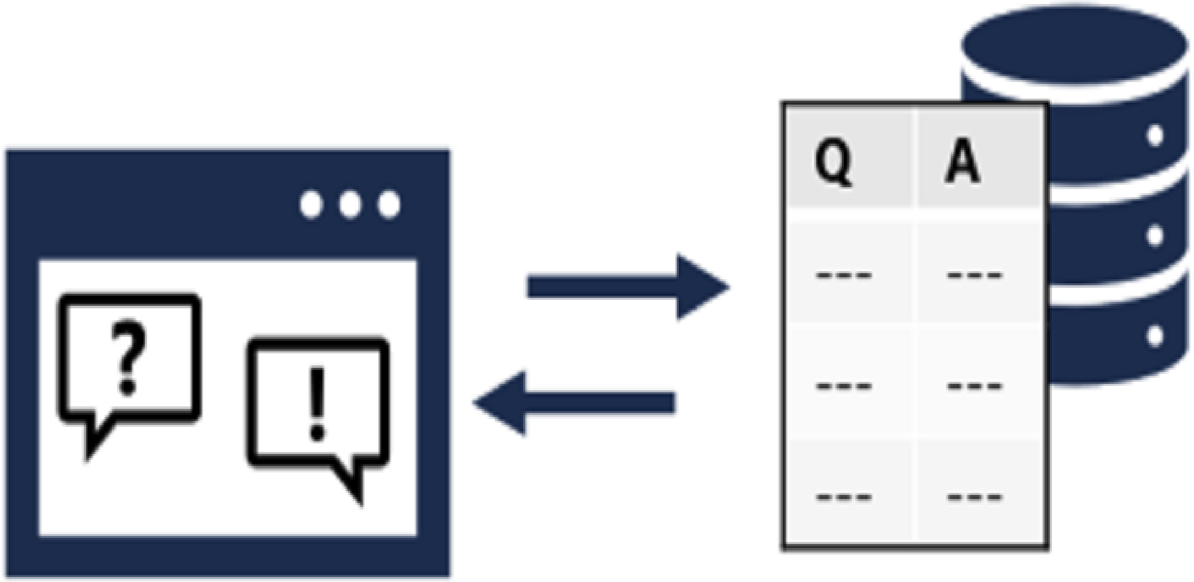
WHAT YOU WILL DO

Lab 07: Objectives

Design custom language understanding.
Define intents, entities, and utterances.
Plan custom question answering.
Create a multilingual question answering strategy.

CUSTOM LANGUAGE

Lab 07: Imported Reference Diagram



How to use it

Connect the diagram to the lab scenario: The support assistant must recognize customer intents and answer frequently asked questions from approved sour...
Use it as a reference while completing the design steps.
Imported from source PPT slide 149.

FOLLOW IN ORDER

Lab 07: Step Sequence

1. Define Intents and Entities
2. Plan Training Data
3. Plan Custom Question Answering
4. Multilingual Strategy
5. Backup and Recovery

CHECK BEFORE MOVING ON

Lab 07: Validation and Checkpoints

You should have intent/entity table, training plan, question answering plan, multilingual strategy, and backup notes.
What is an intent?
What is an entity?
Why add alternate phrasing?
Why should language projects be backed up?

LAB OVERVIEW

Lab 08: Azure AI Search, Knowledge Mining, Vector Search

Main topic: Search and mining
Scenario: The company wants searchable support documents, enriched with extracted text, entities, key phrases, and vector embeddings for RAG.
Validation: You should have index design, data source/indexer plan, skillset plan, query examples, and knowledge store notes.

WHAT YOU WILL DO

Lab 08: Objectives

Design an Azure AI Search solution.
Define indexes, data sources, indexers, and skillsets.
Explain semantic and vector search.
Plan knowledge store projections.

Lab 08: Imported Reference Diagram

How to use it

Connect the diagram to the lab scenario: The company wants searchable support documents, enriched with extracted text, entities, key phrases, and vecto...
Use it as a reference while completing the design steps.
Imported from source PPT slide 380.

FOLLOW IN ORDER

Lab 08: Step Sequence

1. Design an Index
2. Plan Data Sources and Indexers
3. Define a Skillset
4. Query Design
5. Knowledge Store Projections

CHECK BEFORE MOVING ON

Lab 08: Validation and Checkpoints

You should have index design, data source/indexer plan, skillset plan, query examples, and knowledge store notes.
What is an indexer?
What is a skillset?
What is vector search?
Why use semantic ranking?

LAB OVERVIEW

Lab 09: Document Intelligence and Content Understanding

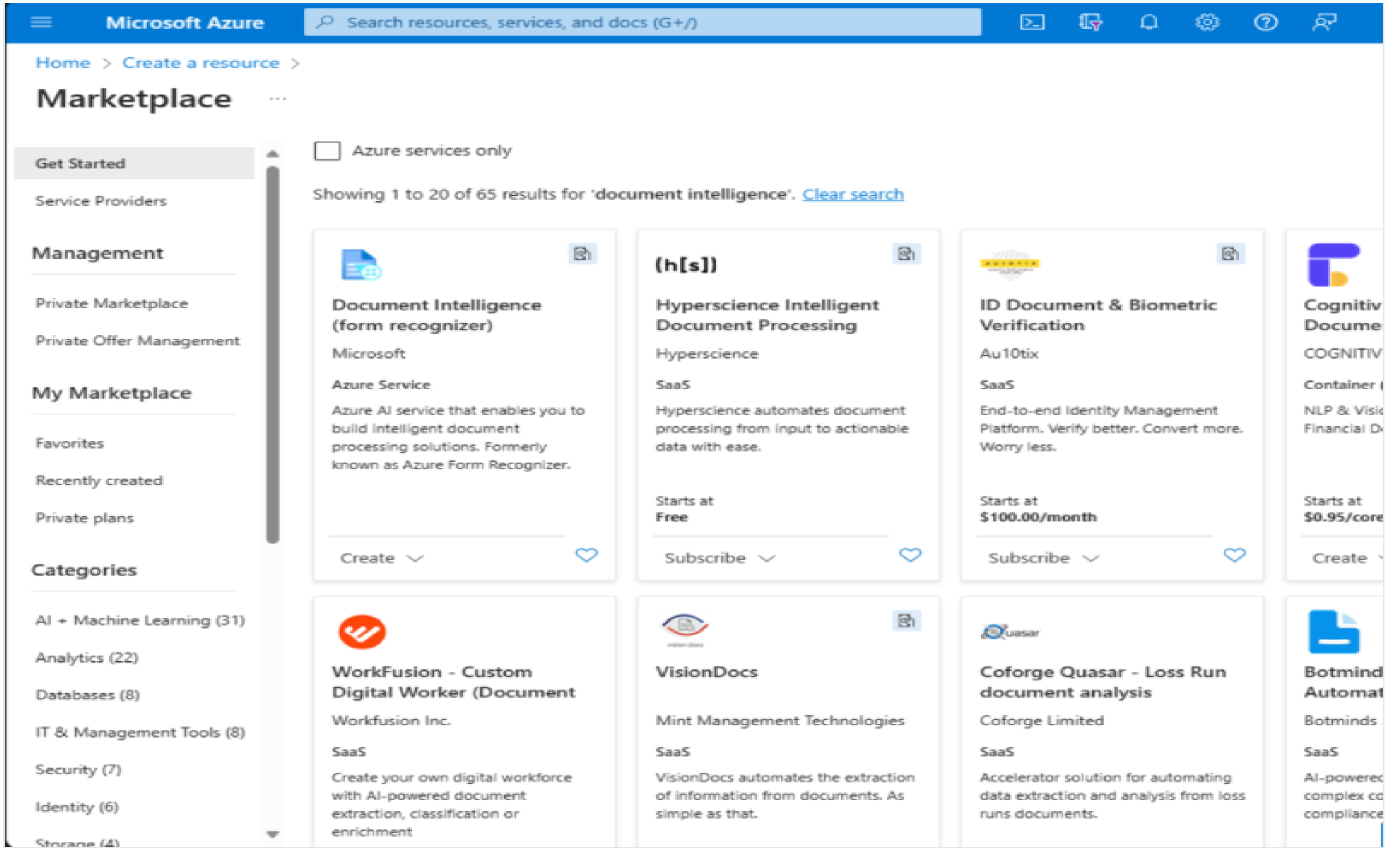
Main topic: Document AI
Scenario: The company processes invoices, contracts, support screenshots, audio notes, and product videos. It wants to extract structured information from multiple content types.
Validation: You should have scenario mapping, custom model plan, content understanding plan, review rules, and integration flow.

WHAT YOU WILL DO

Lab 09: Objectives

Use prebuilt document extraction concepts.
Plan custom Document Intelligence models.
Explain composed models.
Design Content Understanding extraction workflows.

Lab 09: Imported Reference Diagram



How to use it

Connect the diagram to the lab scenario: The company processes invoices, contracts, support screenshots, audio notes, and product videos. It wants to e...
Use it as a reference while completing the design steps.
Imported from source PPT slide 532.

FOLLOW IN ORDER

Lab 09: Step Sequence

1. Map Document Scenarios
2. Custom Document Model Plan
3. Content Understanding Plan
4. Human Review Rules
5. Integration Flow

CHECK BEFORE MOVING ON

Lab 09: Validation and Checkpoints

You should have scenario mapping, custom model plan, content understanding plan, review rules, and integration flow.
When should you use a prebuilt model?
What is a composed model?
Why are confidence scores useful?
What content types can Content Understanding process?

LAB OVERVIEW

Lab 10: AI-102 Capstone and Course Review

Main topic: Integrated review
Scenario: You must design a production-ready customer support AI platform that supports grounded chat, document search, sentiment analysis, speech transcription, image OCR, document extraction, and human escalation.
Validation: You should have a service map, architecture diagram, governance checklist, confidence matrix, cleanup plan, and review plan.

WHAT YOU WILL DO

Lab 10: Objectives

Design an end-to-end Azure AI solution.
Map requirements to Microsoft Foundry and Azure AI services.
Include security, monitoring, responsible AI, and cost controls.
Build a personal review plan.

Lab 10: Imported Reference Diagram

How to use it

Connect the diagram to the lab scenario: You must design a production-ready customer support AI platform that supports grounded chat, document search, ...
Use it as a reference while completing the design steps.
Imported from source PPT slide 716.

FOLLOW IN ORDER

Lab 10: Step Sequence

1. Build a Service Map
2. Draw the Architecture
3. Add Governance Controls
4. Rate Skill Confidence
5. Clean Up Resources
6. Create a 7-Day Review Plan

CHECK BEFORE MOVING ON

Lab 10: Validation and Checkpoints

You should have a service map, architecture diagram, governance checklist, confidence matrix, cleanup plan, and review plan.
Which service should power grounded document search?
When should a human review AI output?
What is the difference between RAG and fine-tuning?
Which AI workload do you need to review most?

COURSE CLOSING

Assessment and Wrap-up

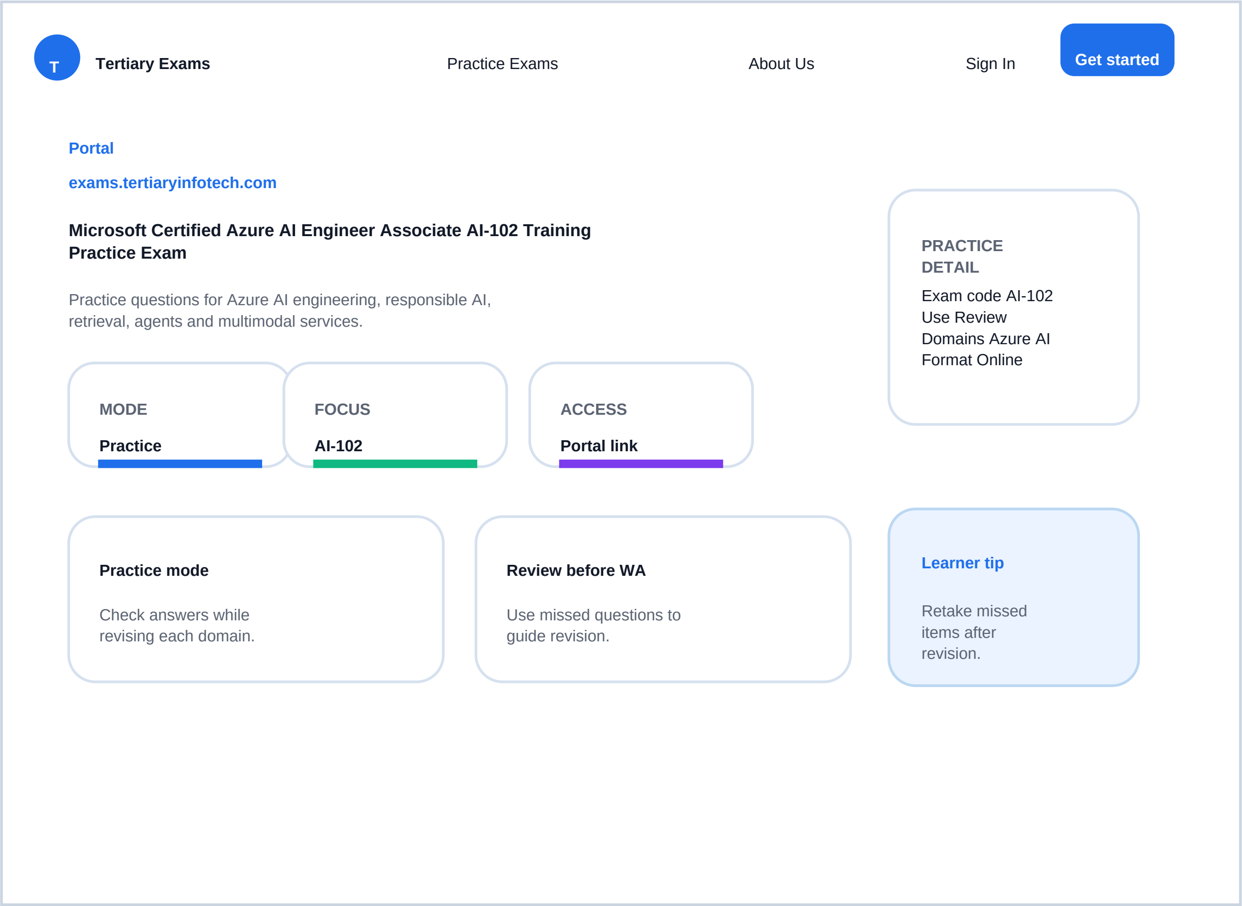
PRACTICE EXAM

Take the Online AI-102 Practice Exam

- Go to the Tertiary Exams portal and open the matching AI-102 Practice Exam.
- Use the practice set to test Azure AI planning, Azure OpenAI, search, vision, language and document intelligence.
- Review mode lets you check the answer and revisit the concept after each question.
- Complete the practice exam before assessment, then bring difficult items to the review session.

Practice exam link:

https://exams.tertiaryinfotech.com



ASSESSMENT

Assessment Flow Reminder

1. **TRAQOM digital attendance**
2. **Assessment digital attendance**
3. **Sit WA then Case Study**
4. **Submit answers on the LMS**
5. **Sign Assessment Summary Record**

The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

BEFORE YOU GO

Certification and TRAQOM Survey

Complete the certification and TRAQOM survey to receive your certificate.
Remember AM, PM and assessment digital attendance.
Contact support if you need help after class.

AFTER CLASS

Support

Email: enquiry@tertiaryinfotech.com
Tel: +65 6100 0613
Website: www.tertiarycourses.com.sg

CONGRATULATIONS

Thank You

You have designed an end-to-end Azure AI engineering solution.